The Wang Method Series

Notebook #5 (Version 1.0)

Structural Classification and State Transition of the 24 Top-Layer Four-Edge-Pair Patterns

A Structural Framework for Understanding Top-Layer Four-Edge-Pair Patterns

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1. Introduction

The four top-layer edge pairs can be arranged in 24 (= 4!) distinct clockwise permutations. Although these permutations appear quite different, they exhibit a surprisingly simple structural organization.

In this notebook, the solved configuration **R B O G** is defined as the **Golden Pattern**.

Rather than studying each permutation individually, we first classify all 24 top-layer four-edge-pair configurations according to the number and relative positions of their aligned edge pairs. The resulting classification consists of six structural groups. We then show that every illegal state can always be transformed into a legal state through one of two deterministic transition paths.

This notebook complements the practical solution methods presented in the previous notebooks by providing a unified structural perspective.

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2. The 24 Top-Layer Four-Edge-Pair Patterns

The 24 possible clockwise permutations are listed in Table 5.1.

Table 5.1. The 24 Top-Layer Four-Edge-Pair Patterns.

Case	Pattern (Clockwise)
1	R B O G
2	R B G O
3	R O B G
4	R O G B
5	R G B O
6	R G O B
7	B R O G
8	B R G O
9	B O R G
10	B O G R
11	B G R O
12	B G O R
13	O R B G
14	O R G B
15	O B R G
16	O B G R
17	O G R B
18	O G B R
19	G R B O
20	G R O B
21	G B R O
22	G B O R
23	G O R B
24	G O B R

Although these 24 patterns appear unrelated, they can be completely classified into only six structural groups.

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3. Structural Classification of All 24 Patterns

Figure 5.1 illustrates the structural classification of all 24 patterns.

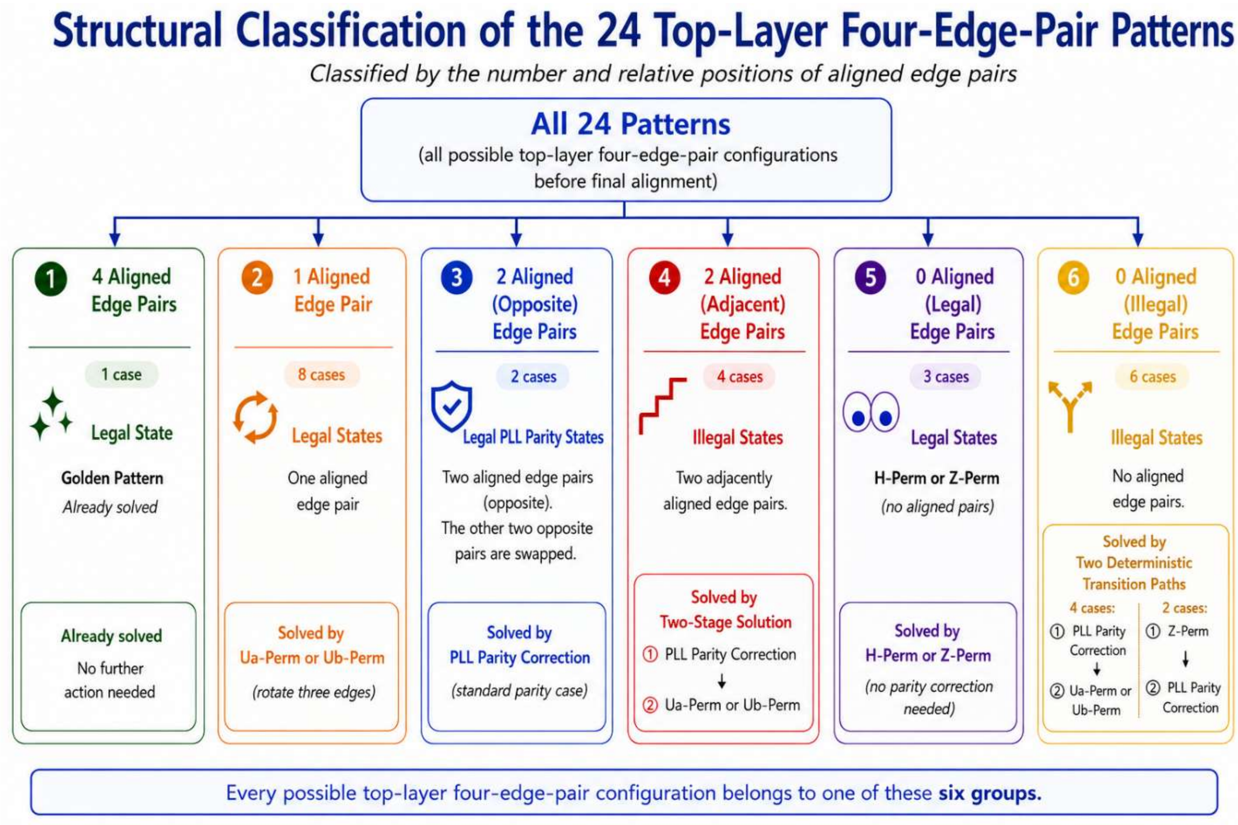

Figure 5.1. Structural classification of the 24 top-layer four-edge-pair patterns before final alignment.

The complete classification and corresponding solution methods are summarized in Table 5.2.

Table 5.2. Classification and Solution of All 24 Patterns.

No. of Aligned Edge Pairs	State Type	Case	Pattern (Clockwise)	Solution
4	Golden (Solved)	1	R B O G	Golden Pattern
1	Legal	4	R O G B	*Ua-Perm*
1	Legal	9	B O R G	*Ua-Perm*
1	Legal	12	B G O R	*Ua-Perm*
1	Legal	16	O B G R	*Ua-Perm*
1	Legal	20	G R O B	*Ua-Perm*
1	Legal	21	G B R O	*Ua-Perm*
1	Legal	5	R G B O	*Ub-Perm*
1	Legal	13	O R B G	*Ub-Perm*
2	Legal (Parity)	6	R G O B	PLL Parity Correction (BG)
2	Legal (Parity)	15	O B R G	PLL Parity Correction (OR)
2	Illegal	2	R B G O	Parity (BO) → R O G B → *Ua-Perm*
2	Illegal	3	R O B G	Parity (OG) → R G B O → *Ua-Perm*
2	Illegal	7	B R O G	Parity (RG) → B G O R → *Ua-Perm*
2	Illegal	22	G B O R	Parity (BR) → G R O B → *Ua-Perm*
0	Legal	17	O G R B	*H-Perm* (OR, GB)
0	Legal	8	B R G O	*Z-Perm* (BR, GO)
0	Legal	24	G O B R	*Z-Perm* (GR, OB)
0	Illegal	11	B G R O	Parity (GO) → B O R G → *Ua-Perm*
0	Illegal	14	O R G B	Parity (RB) → O B G R → *Ua-Perm*
0	Illegal	18	O G B R	Parity (OB) → B G O R → *Ua-Perm*
0	Illegal	23	G O R B	Parity (GR) → R O G B → *Ua-Perm*
0	Illegal	10	B O G R	*Z-Perm* (BR, OG) → R G O B → Parity (GB)
0	Illegal	19	G R B O	*Z-Perm* (GR, BO) → R G O B → Parity (GB)

The 24 patterns are classified into the following six groups.

Group 1 — Golden Pattern

Only one pattern has four aligned edge pairs. This is the solved state and is defined as the Golden Pattern.

Group 2 — One Aligned Edge Pair

Eight patterns have one aligned edge pair. These are all legal states and can be solved directly by either a *Ua-Perm* or a *Ub-Perm*.

Group 3 — Two Opposite Edge Pairs Aligned

Two patterns belong to this group. These are the standard legal PLL parity states and can be solved directly by a standard PLL parity correction.

Group 4 — Two Adjacent Edge Pairs Aligned

Four patterns have two adjacent edge pairs aligned. These are illegal states and require a two-stage solution consisting of a PLL parity correction followed by either a *Ua-Perm* or a *Ub-Perm*.

Group 5 — No Aligned Edge Pairs (Legal States)

Three patterns have no aligned edge pairs but are nevertheless legal states. One pattern is solved by an *H-*Perm, and the other two patterns are solved by a *Z-Perm*.

Group 6 — No Aligned Edge Pairs (Illegal States)

Six patterns have no aligned edge pairs and are illegal states.

Four patterns are transformed into legal *Ua* or *Ub* states through a PLL parity correction, whereas the remaining two patterns first require a *Z-Perm* before reaching legal PLL parity states.

3.1 Key Observations

Several important observations can now be made.

First, the number of aligned edge pairs alone does not determine whether a state is legal.

For example,

- two opposite edge pairs aligned always form legal PLL parity states;
- two adjacent edge pairs aligned always form illegal states;
- no aligned edge pairs may represent either legal or illegal states.

Therefore, both the number and the relative positions of aligned edge pairs are essential for determining legality.

Furthermore, a PLL parity correction may produce either a legal *Ua* state or a legal *Ub* state, depending on which opposite edge pair is selected during the parity correction. These two transition paths are completely symmetric.

Together, these observations partition all 24 top-layer four-edge-pair configurations into six structural groups and provide a unified framework for understanding their legality and solution methods.

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4. Deterministic State Transition of the 10 Illegal States

Among the 24 patterns, 10 are illegal states.

Fortunately, every illegal state can always be transformed into a legal state through one of two deterministic transition paths. Figure 5.2 summarizes these two transition paths.

Eight illegal states are transformed into legal states through a PLL parity correction. Depending on the selected opposite edge pair during the parity correction, the resulting legal state becomes either a *Ua* state or a *Ub* state. The corresponding *Ua-Perm* or *Ub-Perm* then returns the cube to the Golden Pattern.

The remaining two illegal states first require a *Z-Perm*, which converts them into standard legal PLL parity states. A standard PLL parity correction then completes the solution.

Therefore, every illegal state follows one of only two deterministic transition paths.

Most importantly, every illegal state is transformed into a legal state in exactly two stages.

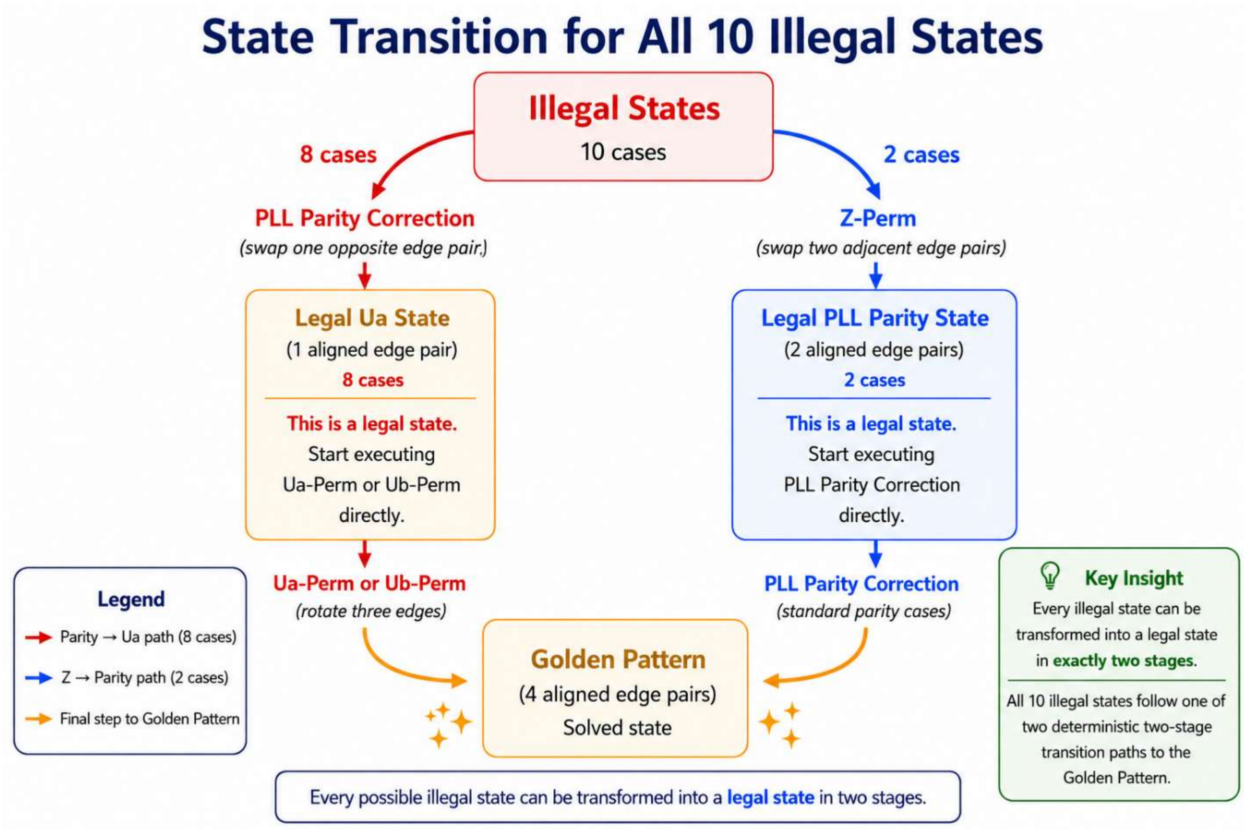

Figure 5.2. State transition of all 10 illegal states

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5. Conclusions

This notebook presents a complete structural classification of all 24 top-layer four-edge-pair patterns.

The major findings are summarized below.

1. The 24 patterns are completely classified into six structural groups.
2. The legality of a pattern depends on both the number and the relative positions of its aligned edge pairs.
3. Two opposite edge pairs aligned always form legal states, whereas two adjacent edge pairs aligned always form illegal states. States with no aligned edge pairs may be either legal or illegal.
4. Every illegal state can always be transformed into a legal state in exactly two deterministic stages.
5. A PLL parity correction may produce either a legal *Ua* state or a legal *Ub* state, depending on the selected opposite edge pair. The two transition paths are completely symmetric.
6. Together, the structural classification and the state-transition framework provide a complete theoretical understanding of the 24 top-layer four-edge-pair patterns and complement the practical solution methods presented in the previous notebooks.

Although different intermediate legal states (*Ua* or *Ub*) may be reached, every illegal state ultimately follows one of two deterministic transition paths back to the Golden Pattern.